

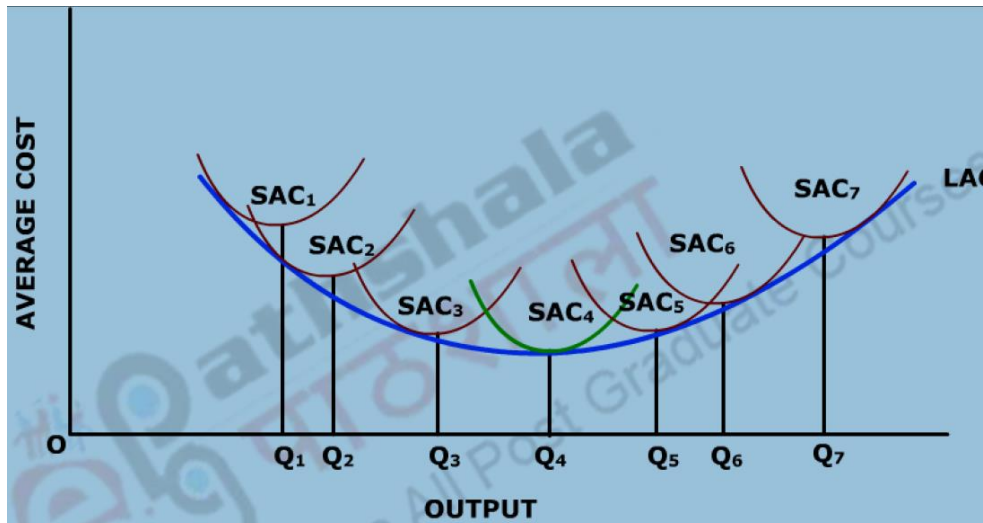
Long Run Costs of Production

In the long-run, all factors of production are variable. All costs are variable costs. The firm can alter the size and scale of the plant to meet the changed market conditions. Therefore, there is LAC curve and LMC curve. Long-run costs are known as PLANNING COSTS as the firm has to plan its future expansion of output. The long-run cost curves are subject to the law of returns to scale as the scale of production may undergo change.

DERIVATION OF THE LONG-RUN COST CURVE:

The Long –run Average cost curve or the LAC curve is the locus of points denoting the least cost of producing different levels of output in the long run. It shows the minimum average costs of producing the corresponding output in the long-run. The LAC curve is thus a planning curve that guides the firm in deciding on the most optimal size of the plant for producing a given level of output. An optimal sized plant is one which enables the production of the output at the minimum costs per unit of output. Given the technology, the firm is free to choose the plant size which entails the least cost. For example, if the firm decides to produce OQ_1 level of output, then it will choose the plant size SAC_2 and not SAC_1 . If the demand for the firm's output increases to OQ_3 , then the average costs starts increasing along the plant SAC_2 and the firm decides to set up a larger plant size SAC_3 to minimize its average costs of production in the long-run. The long-run average curve does not touch the short-run average cost curves on their minimum points. Graphically it can touch the minimum points of SACs only under constant returns to scale. In the phase of increasing returns to scale and decreasing cost, the LAC curve touches the SAC curves to the left of the minimum points of the SAC curves and in the phase of diminishing returns it touches the SAC curves to the right of their lowest point. The LAC is therefore, not the locus of lowest points of SAC curves. The downward sloping portion of LAC comprises of only increasing returns and diminishing cost portions of SAC curves.

The LAC curve is also known as the 'Envelope curve' as it envelops the short run cost curves.



LAC curve (Figure-4)

LONG-RUN MARGINAL COST CURVE (LRMC):

It is the additional cost of producing an additional unit of output in the long-run. Both short run MC curve and long run MC curve refer to the change in total cost due to a unit change in output. The key difference is that LRMC is not attributable to just one or two variable/factor but to all inputs/factors. In the short run, the firm maximizes its profit by equating short run $MR = MC$. And in the long run, the firm maximizes profit by equating long run $MR = MC$. Long run marginal cost shows the change in total cost consequent upon a small change in total output when the firm has adequate time to achieve the output changes by making the appropriate changes in the inputs/factors. The only difference between SMC and LRMC is that while in the short run the existing plant size is continued but in the long run the plant itself is changed due change in output.

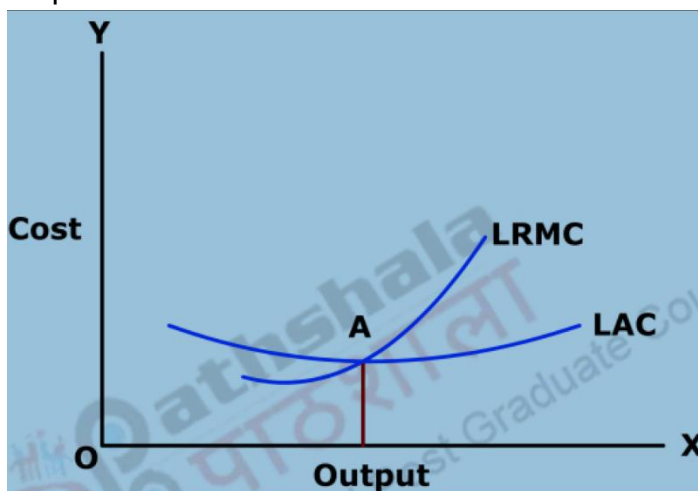


FIGURE 5: LONG RUN AVERAGE COST AND LONG RUN MARGINAL COAT CURVE

Since there are no fixed factor in the long-run, the law of diminishing marginal returns are irrelevant to the LRMC. In the long –run, marginal cost is affected by returns to scale- economies and diseconomies of scale.

RELATIONSHIP BETWEEN LAC and LRMC

The relationship between LAC and LRMC is the same as it exists between the short run average cost SAC and the SMC. LRMC lies below the LAC when LAC is falling and above it when LAC is rising. Thus LAC and LRMC intersect when LAC is the minimum

WHY IS THE LAC CURVE 'U' SHAPED?

In the short run, the shape of the average cost curves essentially reflects the returns to a variable factor as determined by the law of variable proportions. According to this law, as increasing amounts of variable factor are added to the fixed factor, then in the initial stages of production they yield increasing returns but eventually they yield diminishing returns. This explains why per unit costs of production tends to fall initially and ultimately rises up when diminishing returns to the variable factor sets in. Thus, the shape of SRAC curve is 'U' shaped.

The shape of the LAC is however explained by the 'returns to scale'. Returns to scale refers to the change in "optimum cost of production" when the scale of the plant is changed and comprises of-

1. Constant returns to scale – when the successive plants have the same optimum cost.
2. Diminishing returns to scale – when the optimum cost increases with an increase in scale.
3. Increasing returns to scale – when the optimum cost decreases with an increase in scale

The term 'returns to scale signifies two things:

1. it reflects the technical relationship between inputs and output.
2. it shows the changes in cost of production due to non-technical reasons also

These two traits manifest themselves in the form of economies & diseconomies of scale. The 'U' shape of the LAC curve is explained by the economies and diseconomies of scale. Economies means lower per unit cost as output increases and diseconomies is higher per unit cost as output increases.

Economies can be internal and external. Internal economies arise on account of expansion of the firm itself. Internal economies arises due to specialization, choice of more suitable inputs, choice of technology, benefits of large scale production, managerial and supervisory economies and so on. Internal diseconomies arise from exhaustion, difficulties in management, lack of accountability and work culture and so on.

External economies arises from external factors and the firm has no role in it. External economies arises due to expansion of technical knowledge, growth of ancillary industry, development of transport facilities, and availability of banking system and so on. External diseconomies arises due to rise in wages, rise in input prices, pollution and so on.

In case of constant returns to scale, the LAC curve is parallel to the X- axis. The LAC curve is upward sloping in case of diminishing returns to scale and downward sloping in case of increasing returns to scale. Thus, LAC curve is initially downward sloping, parallel to the X-axis up to a point and then upward sloping. The returns to scale determine the shape of the LAC, given the external economies.

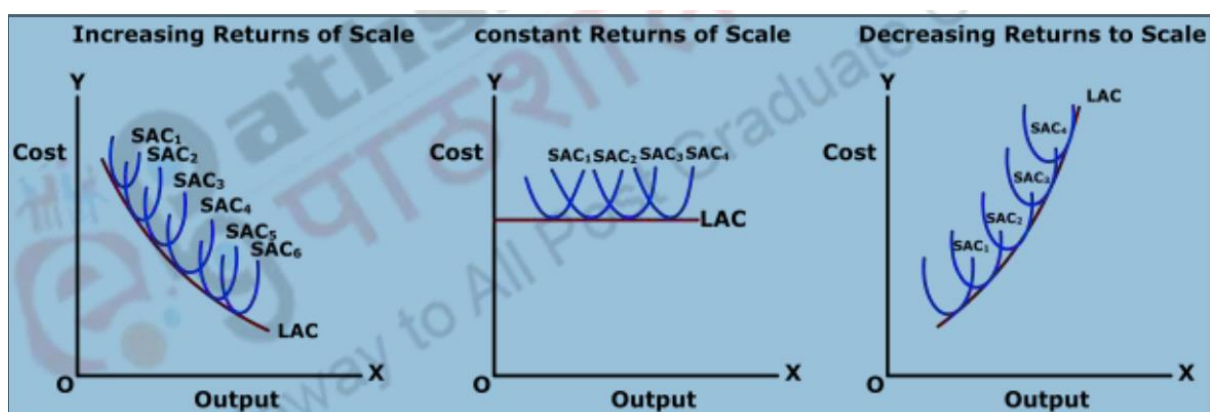


FIGURE 7: DIFFERENT PHASES OF RETURNS TO SCALE

Summary:

The traditional theory of cost believes in a 'U' shaped Short run and Long run cost curves. In short run the U shaped is explain by the law of variable proportion. On the other hand, the laws of returns to scale explains the 'U' in the long run. Along with this the occurrence of both of economies and diseconomies of scale also explain the shape of long run cost curve.

Short run and long run cost and output decision:

In the short run when total cost exceeds total revenue, but revenue exceed variable cost ($P > AVC$), firm suffers losses, but firm continues operating in the short run, but exists industry in the long-run.

If revenues are less than variable costs ($P < AVC$), firm suffers losses. Total losses exceed fixed costs. To minimize firm shuts down.

In the long Run firm will produce a level where LAC is minimum and

$LAC = LMC = SAC = SMC$, will produce at optimum level to get maximum benefit.